

6(a)	(they are) large / AW (molecules / size) ; I large volume A composed of, many / AW, atoms bod large, (molecular) mass / molecular weight lipid not composed of (same or similar), repeating / repeated, subunits / monomers / residues ; ora accept 'many of the, same / similar' for repeating bod units for subunits A protein composed of (monomers of) amino acids A polysaccharide composed of, (monomers of) monosaccharides	2																								
6(b)(i)	peptide bond column correct ; amylopectin and cellulose with glycosidic bond ; glycogen with glycosidic bond <u>and</u> triglyceride with ester bond ; <table><tr><th>biological molecule</th><th>peptide bond</th><th>glycosidic bond</th><th>ester bond</th></tr><tr><td>amylopectin</td><td></td><td>✓</td><td></td></tr><tr><td>cellulose</td><td></td><td>✓</td><td></td></tr><tr><td>collagen</td><td>✓</td><td></td><td></td></tr><tr><td>glycogen</td><td></td><td>✓</td><td></td></tr><tr><td>triglyceride</td><td></td><td></td><td>✓</td></tr></table>	biological molecule	peptide bond	glycosidic bond	ester bond	amylopectin		✓		cellulose		✓		collagen	✓			glycogen		✓		triglyceride			✓	3
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Question	Answer	Marks
6(b)(ii)	amylopectin and glycogen ; <i>both have</i> (monomers of) <u>alpha</u> / α- , glucose or α-1, 4 / α-1, 6 / 1, 4 <u>and</u> 1, 6 (glycosidic) bonds or branched / branching (structure) ; <i>if giving more than one feature then all must be correct, so</i> R beta- / β-, glucose R have <u>only</u>, α-1, 4 bonds / α-1, 6 bonds R coiled / helical I ref. to ring structure for glucose <i>if cellulose with, amylopectin or glycogen, stated, allow one mark if ref. to 1, 4 glycosidic bonds or composed of glucose monomers or are glucose polymers</i>	2
6(c)	either mucous glands / mucous gland cell ; goblet cells ; I ciliated epithelium or <i>two from:</i> and trachea, bronchus, (larger) bronchiole ;; A windpipe R tracheae R bronchio } either order	2

Question	Answer	Marks
6(d)	stimulates / triggers / initiates / AW, an <u>immune response</u> ; causes / AW, production / synthesis, of (specific) antibody against it ; A forms a complex with / binds to / complementary to, an antibody A binds to antigen binding site on antibody <i>plus max two from:</i> high molecular, mass / weight ; activates / triggers a response in, T-lymphocytes / B-, lymphocytes / macrophages / neutrophils / phagocytes A binds to receptors on (T-/B-) lymphocytes <i>also mp2 if binding to, antibody / immunoglobulin, on B-lymphocytes</i> located on the surface of cells / (can) acts as cell marker / involved in <u>cell recognition</u> ; A surface of, pathogens / bacteria / virus A recognised by cells of the immune system A named cell AVP ; e.g. complex structure / AW	3